

MATH3026/4022 Lecture 4

Further Autoregressive Processes

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5th February 2026

- ▶ Properties of the $AR(p)$ process.
- ▶ The Yule-Walker equations for estimation in autoregressive models.
- ▶ Properties of the $AR(2)$ process in detail, including:
 - ▶ Stationarity
 - ▶ Variance
 - ▶ Autocorrelation function

Introduction

During Lecture 3, we defined the **autoregressive (AR) process** and explored some properties of the $AR(1)$ process.

In this lecture, we begin by considering the more general $AR(p)$ process and some key properties.

Recall that the $AR(p)$ process is written

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} + Z_t$$

where Z_t is white noise with mean zero and variance σ_z^2 .

Autoregressive Processes

Rearranging the $AR(p)$ process, we see that

$$\begin{aligned}Z_t &= X_t - \phi_1 X_{t-1} - \dots - \phi_p X_{t-p} \\&= (1 - \phi_1 B - \dots - \phi_p B^p) X_t \\&= \phi(B) X_t\end{aligned}$$

where

$$\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p.$$

Hence, we deduce

$$X_t = [\phi(B)]^{-1} Z_t.$$

$AR(p)$ Process

It can be shown that an $AR(p)$ process written in the compact form

$$\begin{aligned} X_t &= [\phi(B)]^{-1} Z_t \\ &= [1 - \phi_1 B + \dots - \phi_p B^p]^{-1} Z_t \end{aligned}$$

is (weakly) stationary if the roots of the equation

$$\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p = 0$$

lie outside the unit circle.

Here we regard z as a complex variable, noting that the roots of the above equation may be complex numbers.

$AR(p)$ Process

The $AR(p)$ process in compact form can be considered as follows

$$\begin{aligned} X_t &= [\phi(B)]^{-1} Z_t \\ &= \psi(B)Z_t \quad \text{where } \psi(B) = [\phi(B)]^{-1}. \end{aligned}$$

We note that $\psi(B)$ is an infinite sum of the form

$$\begin{aligned} \psi(B) &= (1 - \phi_1 B - \dots - \phi_p B^p)^{-1} \\ &= 1 + \beta_1 B + \beta_2 B^2 + \beta_3 B^3 + \dots \end{aligned}$$

for parameters $\beta_1, \beta_2, \dots$ (derivation not shown) and hence the process is an MA process of infinite order. That is

$$X_t = \sum_{j=0}^{\infty} \beta_j Z_{t-j} \quad (\text{with } \beta_0 = 1).$$

This is a **general linear process** – see Lecture 3.

$AR(p)$ Process

We see that an $AR(p)$ process can be written in the form of a moving average process of infinite order.

Typically, we might refer to this as writing the $AR(p)$ process in $MA(\infty)$ form.

Observing that an $AR(p)$ process may be re-written as an $MA(\infty)$ process makes sense – note that in Lecture 3 we saw that a moving average process can be written as an autoregressive process.

Autoregressive Process

It follows that, for an $AR(p)$ process

$$\mathbb{E}(X_t) = 0 \quad (\text{because } X_t = \sum_{j=0}^{\infty} \beta_j Z_{t-j})$$

$$\text{Var}(X_t) = \sigma_z^2 \sum_{j=0}^{\infty} \beta_j^2 \quad (\text{where } \beta_0 = 1)$$

$$\rho(h) = \frac{\sum_{j=0}^{\infty} \beta_j \beta_{j+h}}{\sum_{j=0}^{\infty} \beta_j^2}.$$

As seen previously in Lecture 3, the process is stationary if $\sum_{j=1}^{\infty} |\beta_j|$ converges.

The Yule-Walker Equations

In practice, attempting to find the coefficients $\{\beta_j\}$ to evaluate the ACF may be difficult.

Instead, we can *assume* that the process is stationary and attempt to find the coefficients.

We followed a similar approach for the $AR(1)$ process during Lecture 3.

The Yule-Walker Equations

Note that

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + Z_t.$$

Multiplying both sides by X_{t-h} ($h > 0$) and taking expectations, we obtain

$$\begin{aligned}\mathbb{E}(X_t X_{t-h}) &= \phi_1 \mathbb{E}(X_{t-1} X_{t-h}) + \dots \\ &\quad + \phi_p \mathbb{E}(X_{t-p} X_{t-h}) + \mathbb{E}(Z_t X_{t-h}) \\ \implies \gamma(-h) &= \phi_1 \gamma(-h+1) + \dots + \phi_p \gamma(-h+p). \quad (4.1)\end{aligned}$$

In the above, we note that $\mathbb{E}(Z_t X_{t-h}) = 0$ (where $h > 0$) because of the causality assumption introduced in Lecture 3 (i.e. current white noise terms are independent of past observations of the series).

The Yule-Walker Equations

Noting that $\gamma(-h) = \gamma(h)$ (stationarity) we deduce that

$$\gamma(h) = \phi_1\gamma(h-1) + \dots + \phi_p\gamma(h-p) \quad \text{for all } h > 0.$$

Dividing the above by $\gamma(0)$, we obtain

$$\rho(h) = \phi_1\rho(h-1) + \dots + \phi_p\rho(h-p) \quad \text{for all } h > 0.$$

The above are a set of difference equations known as the **Yule-Walker equations**.

The Yule-Walker Equations

The Yule-Walker equations have a general solution given by

$$\rho(h) = C_1 \pi_1^{|h|} + \dots + C_p \pi_p^{|h|}$$

where the set of coefficients $\{\pi_j : j = 1, \dots, p\}$ are roots of the auxiliary equation

$$y^p - \phi_1 y^{p-1} - \dots - \phi_p = 0$$

and $C_1, \dots, C_p$ are constants.

The Yule-Walker Equations

The constants $\{C_j : j = 1, \dots, p\}$ are determined subject to initial conditions $\rho(0) = 1$ and $\rho(h) = \rho(-h)$, together with constraints imposed by the set of equations.

The constraints imply that $\sum_{j=1}^p C_j = 1$ (since $\rho(0) = 1$) and hence it follows that

$$\lim_{h \rightarrow \infty} \rho(h) = 0 \text{ if } |\pi_j| < 1 \text{ for all } j.$$

In fact, the condition $|\pi_j| < 1$ for all j is necessary and sufficient for the process to be stationary.

Autoregressive Process of Order 2

During Lecture 3 we covered various properties of the $AR(1)$ process.

Now, we will consider properties of the autoregressive process of order 2 (the $AR(2)$ process).

Recall that the $AR(2)$ process is written

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

where $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with constant variance σ_z^2 .

From Lecture 3 (and by re-arranging the above equation) we know that the $AR(2)$ process may be written in moving average form as

$$Z_t = (1 - \phi_1 B - \phi_2 B^2)X_t.$$

Autoregressive Process of Order 2

We know that the $AR(2)$ process is (weakly) stationary if the roots of the equation

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 = 0$$

lie outside the unit circle.

This implies that if the roots have magnitude > 1 then the process will be (weakly) stationary.

Since the above equation is quadratic, there will be two roots (possibly repeated). We can write the above equation as

$$1 - \phi_1 z - \phi_2 z^2 = (1 - \lambda_1 z)(1 - \lambda_2 z) = 0 \quad (4.2)$$

where, clearly, the roots are $\frac{1}{\lambda_1}$ and $\frac{1}{\lambda_2}$.

Autoregressive Process of Order 2

For (weak) stationarity we require

$$\left| \frac{1}{\lambda_1} \right| > 1 \quad \text{and} \quad \left| \frac{1}{\lambda_2} \right| > 1 \implies |\lambda_1| < 1 \quad \text{and} \quad |\lambda_2| < 1.$$

From Equation (4.2), we note

$$\begin{aligned} (1 - \lambda_1 z)(1 - \lambda_2 z) &= 1 - \phi_1 z - \phi_2 z^2 \\ \implies 1 - (\lambda_1 + \lambda_2)z + \lambda_1 \lambda_2 z^2 &= 1 - \phi_1 z - \phi_2 z^2. \end{aligned}$$

Hence we obtain the following relationship between roots and coefficients

$$\lambda_1 + \lambda_2 = \phi_1 \quad \text{and} \quad \lambda_1 \lambda_2 = -\phi_2.$$

Autoregressive Process of Order 2

We have seen that an $AR(p)$ process can be written as an $MA(\infty)$ process of the form

$$\begin{aligned} X_t &= [\phi(B)]^{-1} Z_t \\ &= \psi(B) Z_t \\ &= \sum_{j=0}^{\infty} \beta_j B^j Z_t \quad \text{where } \beta_0 = 1. \end{aligned}$$

Establishing stationarity and calculating the ACF is often easier when a process is written in MA form.

Hence, we would like to find the coefficients of the polynomial $\psi(B) = [\phi(B)]^{-1}$ in terms of λ_1 and λ_2 .

We begin by finding the roots of $\phi(B) = 0$, considering cases of real unequal, real repeated and complex roots.

$AR(2)$ process: $\phi(B) = 0$ has real, unequal roots

We assume that $\frac{1}{\lambda_1} \neq \frac{1}{\lambda_2} \implies \lambda_1 \neq \lambda_2$.

Whence

$$\begin{aligned}\phi(B)^{-1} &= \frac{1}{(1 - \lambda_1 B)(1 - \lambda_2 B)} \\ &= \frac{A}{1 - \lambda_1 B} + \frac{C}{1 - \lambda_2 B} \quad (\text{partial fractions})\end{aligned}$$

We'll now solve this for A and C .

$AR(2)$ process: $\phi(B) = 0$ has real, unequal roots

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$AR(2)$ process: $\phi(B) = 0$ has real, unequal roots

We see that, where roots are distinct and real

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

where

$$\psi_j = \frac{\lambda_1^{j+1} - \lambda_2^{j+1}}{\lambda_1 - \lambda_2}.$$

$AR(2)$ process: $\phi(B) = 0$ has real, repeated roots

We assume that $\frac{1}{\lambda_1} = \frac{1}{\lambda_2} = \frac{1}{\lambda} \implies \lambda_1 = \lambda_2 = \lambda$.

Whence

$$\phi(B)^{-1} = \frac{1}{(1 - \lambda B)^2}.$$

Using a Taylor series expansion we can show that

$$\frac{1}{(1 - \lambda B)^2} = \sum_{j=0}^{\infty} (j+1) \lambda^j B^j$$

and therefore, where roots are real and repeated

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j} \quad \text{where } \psi_j = (j+1) \lambda^j.$$

AR(2) process: $\phi(B) = 0$ has complex roots

Complex roots would occur when the discriminant of the equation

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 = 0$$

is less than zero.

Where this is the case, we will obtain two distinct complex roots that are complex conjugates of one another. That is

$$\frac{1}{\lambda_1} \neq \frac{1}{\lambda_2} \quad \text{and} \quad \lambda_1 = \lambda e^{i\theta}, \lambda_2 = \lambda e^{-i\theta}.$$

To obtain the coefficients in the moving average representation of the process we need only use the results from the case of two distinct real roots (covered earlier) but replace λ_1 and λ_2 with the complex numbers given above.

AR(2) process: $\phi(B) = 0$ has complex roots

Hence we see that

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

where

$$\begin{aligned}\psi_j &= \frac{\lambda_1^{j+1} - \lambda_2^{j+1}}{\lambda_1 - \lambda_2} \\&= \frac{(\lambda e^{i\theta})^{j+1} - (\lambda e^{-i\theta})^{j+1}}{\lambda e^{i\theta} - \lambda e^{-i\theta}} \\&= \frac{\lambda^{j+1} (e^{i(j+1)\theta} - e^{-i(j+1)\theta})}{\lambda(e^{i\theta} - e^{-i\theta})} \\&= \frac{\lambda^j (e^{i(j+1)\theta} - e^{-i(j+1)\theta})}{e^{i\theta} - e^{-i\theta}}\end{aligned}$$

$AR(2)$ process: $\phi(B) = 0$ has complex roots

Recall that

$$e^{ik\theta} - e^{-ik\theta} = 2i \sin(k\theta).$$

Then

$$\begin{aligned}\psi_j &= \frac{\lambda^j (2i \sin((j+1)\theta))}{2i \sin \theta} \\ &= \frac{\lambda^j \sin((j+1)\theta)}{\sin \theta}.\end{aligned}$$

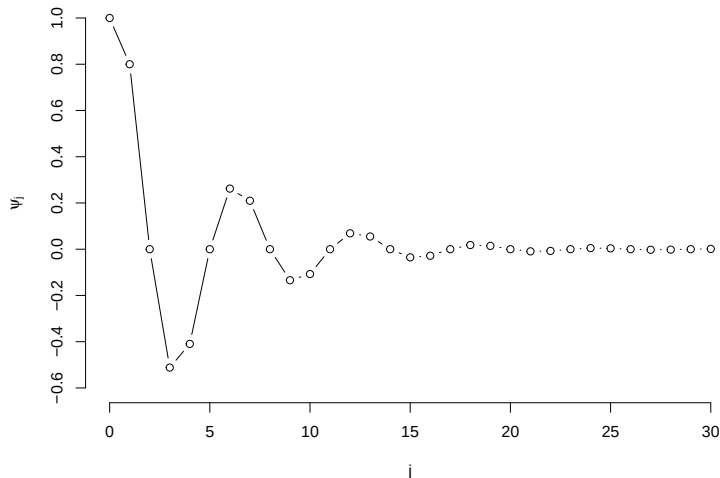
So

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j} \quad \text{with} \quad \psi_j = \frac{\lambda^j \sin((j+1)\theta)}{\sin \theta}.$$

Since $|\frac{1}{\lambda_1}| = |\frac{1}{\lambda_2}| = |\frac{1}{\lambda}| > 1$ it follows that $|\lambda| < 1$ and the coefficients, ψ_j , tend to zero as $j \rightarrow \infty$.

$AR(2)$ process: $\phi(B) = 0$ has complex roots

The plot below shows example coefficients for the $MA(\infty)$ representation of the $AR(2)$ process where $\lambda = 0.8$ and $\theta = \pi/3$.



Autocorrelation function for the $AR(2)$ process

To calculate the ACF for the $AR(2)$ process, we assume that the process is stationary *a priori* and use the Yule-Walker equations, given by

$$\rho(h) = \phi_1\rho(h-1) + \phi_2\rho(h-2).$$

We know that $\rho(0) = 1$ and

$$\begin{aligned}\rho(1) &= \phi_1\rho(0) + \phi_2\rho(-1) \\ &= \phi_1 + \phi_2\rho(1) \quad (\text{since } \rho(1) = \rho(-1))\end{aligned}$$

Rearranging, we see that

$$\rho(1) = \frac{\phi_1}{1 - \phi_2}.$$

Autocorrelation function for the $AR(2)$ process

The solutions to the Yule-Walker equations (a set of difference equations) take the form

$$\rho(h) = C_1 \lambda_1^{|h|} + C_2 \lambda_2^{|h|}$$

where λ_1 and λ_2 are the roots (possibly repeated or complex) of the auxiliary equation

$$y^2 - \phi_1 y - \phi_2 = 0.$$

Next, we'll solve the equations using the boundary conditions from the previous slide, for the cases of real unequal, real repeated and for complex roots.

ACF for the $AR(2)$ process – Real, distinct roots

ACF for the $AR(2)$ process – Real, distinct roots

ACF for the $AR(2)$ process – Real, repeated roots

ACF for the $AR(2)$ process – Complex roots

ACF for the $AR(2)$ process – Complex roots

Variance of the $AR(2)$ process

Now we turn our attention to calculating the variance of the $AR(2)$ process. Recall that the $AR(2)$ process is written

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t. \quad (4.3)$$

Multiplying both sides of the above by Z_t and taking expectations, we obtain

$$\begin{aligned} \mathbb{E}(X_t Z_t) &= \phi_1 \mathbb{E}(X_{t-1} Z_t) + \phi_2 \mathbb{E}(X_{t-2} Z_t) + \mathbb{E}(Z_t^2) \\ &= \mathbb{E}(Z_t^2) \\ &= \sigma_z^2. \end{aligned}$$

Since future values of Z_t are assumed to be independent of past values of X_t (i.e. $\mathbb{E}(X_{t-1} Z_t) = \mathbb{E}(X_{t-2} Z_t) = 0$).

Variance of the $AR(2)$ process

Now, we multiply both sides of (4.3) by X_t and take expectations, whence

$$\begin{aligned}\mathbb{E}(X_t^2) &= \phi_1 \mathbb{E}(X_t X_{t-1}) + \phi_2 \mathbb{E}(X_t X_{t-2}) + \mathbb{E}(X_t Z_t) \\ \implies \text{Var}(X_t) &= \phi_1 \gamma(1) + \phi_2 \gamma(2) + \sigma_z^2.\end{aligned}$$

We note that $\text{Var}(X_t) = \gamma(0)$ and, hence, it follows that

$$\begin{aligned}\gamma(0) &= \phi_1 \gamma(1) + \phi_2 \gamma(2) + \sigma_z^2 \\ \implies \sigma_z^2 &= \gamma(0) - \phi_1 \gamma(1) - \phi_2 \gamma(2) \\ \implies \sigma_z^2 &= \gamma(0) \left[1 - \phi_1 \frac{\gamma(1)}{\gamma(0)} - \phi_2 \frac{\gamma(2)}{\gamma(0)} \right] \\ &= \gamma(0) [1 - \phi_1 \rho(1) - \phi_2 \rho(2)].\end{aligned}$$

Hence

$$\text{Var}(X_t) = \gamma(0) = \frac{\sigma_z^2}{1 - \phi_1 \rho(1) - \phi_2 \rho(2)}.$$

$AR(2)$ process: Stationarity Conditions

We derive some stationarity conditions for the $AR(2)$ process (in terms of ϕ_1 and ϕ_2).

$AR(2)$ process: Stationarity Conditions

$AR(2)$ process: Stationarity Conditions

To summarise, stationarity conditions for the $AR(2)$ process are

$$\phi_1 + \phi_2 < 1$$

$$\phi_2 - \phi_1 < 1$$

$$|\phi_2| < 1$$

And we may express the set of values of (ϕ_1, ϕ_2) , for which these conditions hold, graphically.

$AR(2)$ process: Stationarity Conditions - Graphical Representation

Lecture 4: Learning Outcomes

- ▶ Know how to define an $AR(p)$ process and understand properties of this process.
- ▶ Understand what is meant by the Yule-Walker equations and be able to solve these equations for some time series.
- ▶ Properties of the $AR(2)$ process including stationarity conditions and calculation of the variance and ACF.